

Principles of Information Security

Dr. Ashok Kumar Das

Center for Security, Theory and Algorithmic Research
International Institute of Information Technology, Hyderabad

E-mail: ashok.das@iiit.ac.in

URL: <http://www.iiit.ac.in/people/faculty/ashokkdas>

Personal Home Page: <http://sites.google.com/site/iitkgpakdas/>

Signature Schemes

- A *signature scheme* is a five-tuple $(\mathcal{P}, \mathcal{A}, \mathcal{K}, \mathcal{S}, \mathcal{V})$, where the following conditions are satisfied:
 - 1. $\mathcal{P}$ is a finite set of possible messages;
 - 2. $\mathcal{A}$ is a finite set of possible signatures;
 - 3. $\mathcal{K}$, the keyspace, is a finite set of possible keys;
 - 4. For each $k \in \mathcal{K}$, there is a signing algorithm $sig_k \in \mathcal{S}$ and a corresponding verification algorithm $ver_k \in \mathcal{V}$. Each $sig_k : \mathcal{P} \rightarrow \mathcal{A}$ and $ver_k : \mathcal{P} \times \mathcal{A} \rightarrow \{true, false\}$ are functions such that the following equation is satisfied for every message $x \in \mathcal{P}$ and for every signature $y \in \mathcal{A}$:
$$ver_k(x, y) = true, \text{ if } y = sig_k(x),$$
$$ver_k(x, y) = false, \text{ if } y \neq sig_k(x).$$
- The pair (x, y) with $x \in \mathcal{P}$ and $y \in \mathcal{A}$ is called a *signed message*.

The Digital Signature Algorithm (DSA)

- The DSA is based on the difficulty of computing logarithms and is based on schemes originally presented by ElGamal and Schnorr.

Table: Global Public-Key Components

p	prime number where $2^{L-1} < p < 2^L$ for $512 \leq L \leq 1024$ and L is a multiple of 64.
q	prime divisor of $(p - 1)$, where $2^{159} < q < 2^{160}$; i.e., bit length of 160 bits.
g	$= h^{(p-1)/q} \bmod p$, where h is any integer with $1 < h < p - 1$ such that $h^{(p-1)/q} \bmod p > 1$.

The Digital Signature Algorithm (DSA) (Continued...)

Table: User's Private Key

x	random or psuedo-random integer with $0 < x < q$.
-----	--

Table: User's Public Key

$y = g^x \bmod p.$

Table: User's Per-Message Secret Number

k	random or psuedo-random integer with $0 < k < q$.
-----	--

The Digital Signature Algorithm (DSA) (Continued...)

Table: Signing Phase

$r = (g^k \bmod p) \bmod q$ $s = [k^{-1}(H(M) + x.r)] \bmod q$ $\text{Signature} = (r, s)$ $\text{Send } (M, (r, s)) \text{ to reviewer.}$
--

M : message to be signed

Digital Signatures

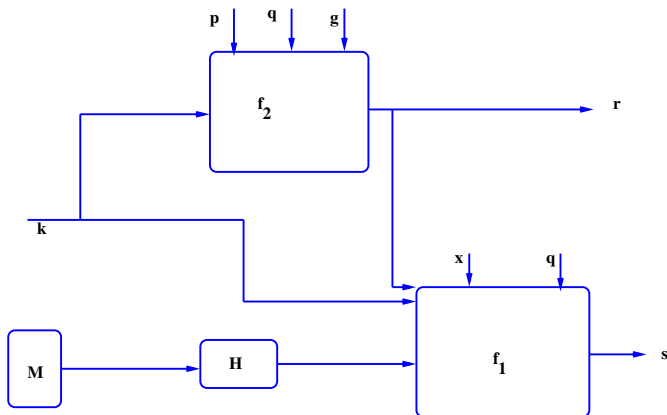


Figure: (a) Signing

$$s = f_1(H(M), k, x, r, q) = [k^{-1}(H(M) + x.r)] \bmod q$$
$$r = f_2(k, p, q, g) = (g^k \bmod p) \bmod q$$

The Digital Signature Algorithm (DSA) (Continued...)

Table: Verification Phase

$$w = (s')^{-1} \bmod q$$

$$u1 = [H(M').w] \bmod q$$

$$u2 = (r').w \bmod q$$

$$v = (g^{u1} \cdot y^{u2} \bmod p) \bmod q$$

TEST: $v = r'$. If so accept; otherwise reject.

M', r', s' : received versions of M, r, s

Digital Signatures

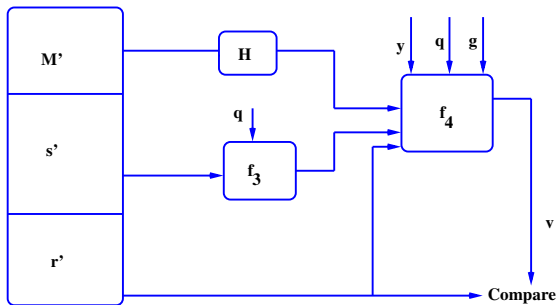


Figure: (b) Verifying

$$\begin{aligned} w &= f_3(s', q) = (s')^{-1} \bmod q \\ v &= f_4(y, q, g, H(M'), w, r') \\ &= ((g^{(H(M') \cdot w) \bmod p}) \cdot y^{(r' \cdot w) \bmod q}) \bmod p \bmod q \end{aligned}$$

The RSA Digital Signature

- The The RSA Digital Signature algorithm is based on the difficulty of factoring a composite number into two prime factors.

Table: Key generation of the RSA digital signature algorithm

Select p, q	p and q both prime, $p \neq q$ (p and q are large)
Calculate $n = p \times q$	
Calculate $\phi(n) = (p - 1)(q - 1)$	
Select integer e	$\gcd(e, \phi(n)) = 1; 1 < e < \phi(n)$
Calculate d	$d \equiv e^{-1} \pmod{\phi(n)}$
Public key	$KU = \{e, n\}$
Private key	$KR = \{d, n\}$

The RSA Digital Signature (Continued...)

Setup Phase

- $n = p \times q$
- $\mathcal{P} = \mathcal{A} = \mathcal{Z}_n$
- Define the key space as
$$\mathcal{K} = \{(n, p, q, e, d) : n = pq; p, q \text{ primes, } ed \equiv 1 \pmod{\phi(n)}, \gcd(e, \phi(n)) = 1\}$$
- Let two users A and B be Alice and Bob, respectively.
- Alice publishes $\{e, n\}$ and keeps $\{d, n\}$ as private.
- $x \in \mathcal{P}$ be a message to be signed by the Alice (User A).

The RSA Digital Signature (Continued...)

Signature Generation Phase

- A computes $y = \text{sig}_K(x) = x^d \pmod{n}$.
- The signature on the message x is y .
- The signed message on x is (x, y) .
- $Alice \rightarrow Bob : (x, y)$.

The RSA Digital Signature (Continued...)

Signature Verification Phase

- B computes $x' = y^e \pmod n$.
- Bob accepts signature as valid if $x' = x$.
- Bob otherwise rejects the signature.

The RSA Digital Signature (Continued...)

Correctness Proof of Signature Verification Phase

$$\begin{aligned}x' &= y^e \pmod n \\&= (x^d \pmod n)^e \pmod n \\&= x^{ed} \pmod n, \text{ since } ed \equiv 1 \pmod{\phi(n)} \\&= x^{k\phi(n)+1} \pmod n \\&= x.\end{aligned}$$